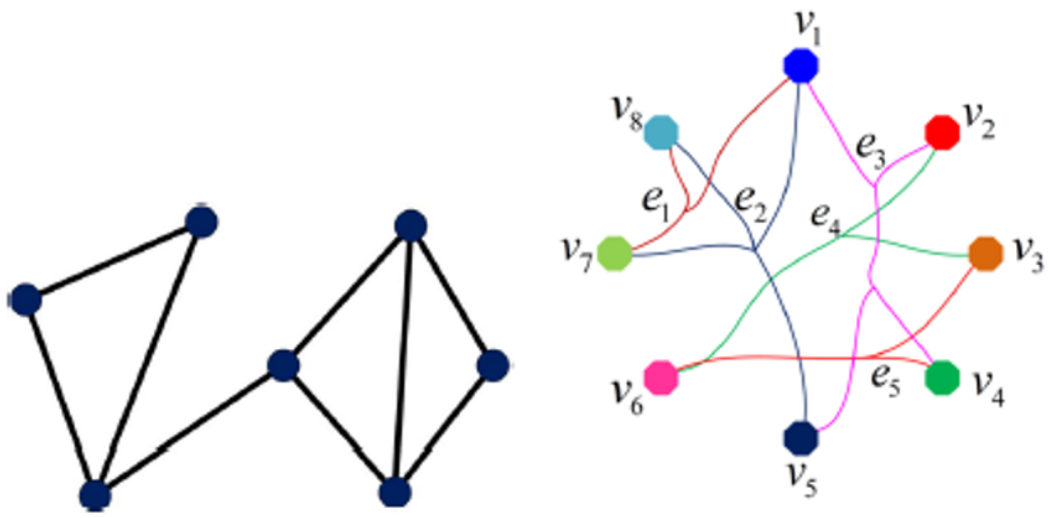
Tsinghua Shenzhen International Graduate School
Data Visualization
Spring 2024

Mid-Term Assignment

Issued: April 2, 2024 **Due:** April 20, 2024

In the mid-term assignment, you will have the opportunity to apply the techniques you have learned to create a integrative visualization diagram about hypergraph with your teammates.

About Hypergraph



(a) A graph (b) A hypergraph

A hypergraph, comprising a set of vertices and hyperedges, represents a generalization of a traditional graph. As depicted in the accompanying figure, a key distinction between these two structures is that while simple graphs feature edges that connect only two vertices, hyperedges in a hypergraph can connect any number of vertices. Consequently, the edge degree in a hypergraph can be significantly greater than the fixed two-degree edges found in simple graphs. This structural difference allows hypergraphs to excel in modeling complex correlations present in real-world data, which often extend beyond simple pairwise relationships. In contrast, traditional graph structures are limited to modeling pairwise connections due to their two-degree edges.

To further enhance your understanding of hypergraphs and their potential applications, we encourage you to explore the following survey paper:

`https://ieeexplore.ieee.org/abstract/document/9264674`

Requirements

1.1. To begin this visualization assignment, it is essential that you first engage in a thorough reading of the hypergraph survey paper provided. This will ensure that you have a solid understanding of the key concepts, theories, and applications related to hypergraphs.

1.2. Your group is required to build an **interactive** visualization project **about hypergraph**. Please carefully review the following instructions and guidelines:

(a) Your visualization project must include **an interactive module**, allowing users to engage with the visualization using their mouse or keyboard for a more immersive experience.

(b) Your visualization should **center around "hypergraph"**. This means that you can either design an interactive platform to help users gain a deeper understanding of hypergraph itself or employ hypergraph as a visualization tool to display information from other datasets. Drawing from our previous course experience, please ensure that your project's focus is on "hypergraph" rather than traditional "graph".

If your group is still unsure about how to approach the project or how to achieve the desired interactivity, consider the following three example ideas for inspiration:

i. Demonstrate the dynamic algorithm processes involved in hypergraph applications, such as clustering and other problem-solving methods. Enable users to customize the training data, while the visualization dynamically displays each step of the algorithm in action.

ii. Facilitate a dynamic comparison of the similarities and differences between hypergraphs and traditional graphs. Allow users to add, delete, modify, and query both hypergraph and graph structures simultaneously. By visually contrasting the two, the dynamic visualization can highlight the benefits of hypergraphs.

iii. Utilize hypergraph as a mean to visualize the collaboration and competition among players in a football match dataset. Design an interactive hypergraph that enables users to focus on specific star players and analyze their performance data in detail.

Remember, these are just example ideas to spark your creativity. Feel free to explore other innovative approaches to visualize about hypergraph and showcase their potential applications in an interactive manner.

(c) You are not restricted to specific visualization tools for this assignment. However, we strongly recommend utilizing dynamic web coding with HTML and JavaScript to create and present your visualization work. Here's an example of a dynamic web-based visualization for your reference:

`https://distill.pub/2021/gnn-intro/`

Additionally, we suggest taking the course "CS50 Web Programming with Python and JavaScript 2020" to enhance your web coding skills. You can find the course videos on the Bilibili platform at the following link:

`https://www.bilibili.com/video/BV1s64y1e7gp/?share_source=copy_web&vd_source=d8726e08c49ad831f235c68fc1097632`

1.3. Your group is required to write a **report** with a description and interpretation of your project. The report should contain, at minimum, the following information:

(a) An introduction to your project.

(b) If you have incorporated a new dataset into your assignment, please provide a comprehensive description of the dataset, including its attributes, types, and any other relevant information.

(c) Several screenshots showcasing the visualization project you have created for this assignment.

(d) Detailed instructions on how to interact with your visualization project. In this section, explain all the functionalities of your visualization and provide guidance on how to utilize them effectively.

(e) An explanation of the visual encoding and interactivity techniques utilized in your project. This section should detail the methods you have employed to represent your data visually and to facilitate user interaction, discussing the choices of colors, sizes, shapes, and other design elements, as well as the various interactive features that contribute to the overall effectiveness and aesthetics of your visualization.

1.4. It is important to be transparent about any external resources used in completing an assignment. If your group has used any large language models, such as ChatGPT, Github Copilot, or BLOOM, please specify how you used the external resource(s) - for example, did you use them to brainstorm ideas, write code, or proofread your work? If your group has completed the homework unaided, please state that explicitly in your report by saying "We have completed this homework unaided."

Presentation

Your group is required to prepare for an **oral presentation** to showcase your work during the class sessions on **4.24**. Please adhere to the following instructions:

1.1. The presentation will be conducted on a group basis. However, within each group, every member must give a presentation to introduce their **individual contributions** to the overall project.

1.2. Each member's presentation should not exceed **3 minutes**. After all members have presented, there will be a **3 minutes'** Q&A session for the entire group. The total presentation time for each group will depend on the number of group members. For example, a group of 6 students will have a total presentation time of 3 * 6 + 3 = 21 minutes. Please practice your presentation to ensure that you can effectively convey all necessary information within this time frame.

1.3. Your group should prepare a **slide** to accompany the presentation. This slide deck should incorporate the content from all group members' presentations, allowing for a smooth transition between each member's segment.

1.4. The order of the presentations will be determined by a random draw.

Submission

To ensure that your group's submission is complete and easy to manage, please make sure to follow these guidelines:

1.1. Your group's submission should consist of two parts: **the report** and **the appendix**. Please compress both parts into a single **.zip** format file, which should be named using the following format: 'Group-[Your Group Number].zip'.

1.2. The report should be integrated into a single PDF file, named 'report.pdf'. Please make sure to put this file outside of the appendix part folder, so it's easy to locate.

1.3. The appendix should be placed in a folder named "Visualization Appendix". This folder should contain **all of your code**, **all of your group member's slides**, and other appendix files.

1.4. To facilitate seamless uploading and downloading, please ensure that the size of your submitted file does not exceed 50MB. If your file size surpasses the upload limit, kindly reach out to the teaching assistant for additional guidance on how to proceed.

Grading

The mid-term assignment will be evaluated based on your group's presentation and report. The grading will take into consideration the following criteria:

- Completeness (40%): Have your group completed the visualization tasks using your chosen visualization tools? Have your group provided a description and interpretation for your topic and project in the report? Have your group successfully presented your work in a clear and concise manner?
- Clarity(30%): Have your group made your projects easy to read and understand? Have your group used appropriate colors, sizes, shapes, and legends to distinguish different elements on your projects? Have your group added informative titles and text descriptions to your projects? Have your group provided a well-prepared presentation, or given a thoughtful answer to the presentation problem?
- Creativity(30%): Have your group gone beyond the minimum requirements of the assignment to create a particularly engaging design? Have your group used creative approaches to visualize your data and convey your message effectively?